

Lokesh Reddy Elluri

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IEEE-published MS Data Science candidate at Indiana University Bloomington (GPA 3.74, May 2026) focused on production ML systems, transformer-based NLP, and large-scale GenAI pipelines. Proven track record building low-latency inference services, fine-tuning LLMs, and architecting RAG systems with LangChain, FAISS, and FastAPI. Targeting Data Scientist, Machine Learning Engineer, and AI Engineer roles starting May 2026.

EDUCATION

Indiana University Bloomington <i>M.S. in Data Science, GPA: 3.74/4.00</i> Coursework: Applied Machine Learning, Data Mining, Big Data Management, Advanced Databases, Applied Algorithms, Statistics.	Bloomington, IN <i>Aug 2024 – May 2026</i>
Vellore Institute of Technology <i>B.Tech in Computer Science (Bioinformatics), GPA: 8.14/10</i> Coursework: Object-Oriented Programming, Data Structures and Algorithms, Database Management Systems, Statistical Analysis, Software Engineering.	Vellore, India <i>Sep 2020 – May 2024</i>

TECHNICAL SKILLS

Languages:	Python, SQL, Java
Machine Learning:	PyTorch, Scikit-learn, XGBoost, Model Evaluation, Feature Engineering, Hyperparameter Tuning
NLP & LLMs:	Transformers, Hugging Face, RAG, LangChain, Prompt Engineering, Fine-tuning
Data & Systems:	Pandas, NumPy, Spark, ETL Pipelines, MLflow, Airflow
Backend & Deployment:	FastAPI, REST APIs, Docker, AWS (S3, EC2, Lambda), PostgreSQL
Tools:	Git, Linux, FAISS, Streamlit, Jupyter; Data Structures, Algorithms, System Design

EXPERIENCE

Vidal Health – Data Scientist Intern (Feb 2024 – Apr 2024) - Owned an end-to-end NLP intent classification pipeline (TF-IDF + Logistic Regression baseline, fine-tuned DistilBERT in production) serving 5,000+ monthly customer queries at sub-200ms p95 latency, deflecting 15% of L1 support tickets. - Productionized a reproducible training workflow with stratified k-fold cross-validation and Bayesian hyperparameter search, boosting macro F1-score from 0.71 to 0.86 and trimming average handling time by 30% across production traffic. - Pioneered an LLM-assisted weak supervision pipeline for large-scale data annotation, cutting labeling cost by 40% while improving inter-annotator agreement and downstream classifier robustness on edge-case intents. - Shipped Dockerized FastAPI inference services wired into CI/CD, reclaiming roughly 12 engineering hours per week of manual triage and release overhead.	Bengaluru, India
Team Lease – Software Engineer Intern (May 2023 – Jun 2023) - Launched a distributed ATS-integrated recruitment platform on Django, PostgreSQL, and JavaScript serving 500+ concurrent applicants; tuned indexing strategies and query plans, cutting p99 read latency by 35%. - Delivered REST microservices for candidate tracking and shipped real-time analytics dashboards backed by materialized views, compressing time-to-hire by 10% and surfacing hiring funnel insights to non-technical stakeholders. - Drove 80% unit and integration test coverage on critical endpoints, hardening release stability and enabling weekly zero-downtime deployments through automated regression gates.	Bengaluru, India

PROJECTS

RAG Document Q&A System – Python, LangChain, FAISS, OpenAI, Sentence Transformers, FastAPI, Docker - Engineered a production-grade retrieval-augmented generation pipeline with semantic chunking, BGE sentence embeddings, FAISS HNSW vector indexing, and a cross-encoder reranker, achieving sub-2-second p95 query latency on 500-page corpora. - Built a custom evaluation harness measuring retrieval hit@k, answer faithfulness, and context precision across heterogeneous documents; iterated on chunk size and reranker depth to lift grounded-answer accuracy by 22%. - Packaged the full stack with Docker behind FastAPI inference endpoints, enabling reproducible cloud deployment and horizontal scaling behind a Streamlit evaluator UI.	GitHub
Customer Churn Prediction System – Python, Scikit-learn, XGBoost, SHAP, FastAPI, Streamlit - Developed a feature-rich ML pipeline benchmarking Logistic Regression, Random Forest, and gradient-boosted XGBoost on telecom data, reaching 84% ROC-AUC and isolating top churn drivers through SHAP attribution analysis. - Shipped a low-latency FastAPI scoring service and Streamlit dashboard supporting real-time customer scoring, segment-level retention insights, and counterfactual what-if analysis for stakeholders.	GitHub
JPEG Forensic Analysis Tool – Python, NumPy, OpenCV, DCT, Jupyter - Built a DCT-based image forensics pipeline detecting JPEG recompression artifacts via block-level coefficient histogram analysis, enabling tamper localization for digital forensics workflows on real-world image datasets. - Optimized core DCT routines with NumPy vectorization, accelerating per-image analysis 8x over the naive baseline and supporting batch evaluation across thousands of images.	GitHub

PUBLICATION

- Elluri, L., et al.** “Real-time XSS Vulnerability Detection Using Machine Learning,” *IEEE CONIT 2023*. Authored a feature-based ML classifier reaching 82.65% accuracy and 80% recall for proactive web security; indexed in [IEEE Xplore](#).

LEADERSHIP

- Community Development Coordinator**, Graduate and Professional Student Government, Indiana University Bloomington (2025–Present)
- Sergeant at Arms**, Toastmasters International, VIT Chapter (2022).